

Andrew Choy Special Participation A: HW8 Gemini Thinking

Conversation Log: **Lazy Prompt**

November 26, 2025

Interaction 1: SSM Convolution Kernel

User Prompt: *Tackle problem 1 from this pdf*

Solution Derivation

(a) Convolution Kernel and the Output Equation

Goal: Show $y_k = \sum_{l=0}^L K_l u_{k-l}$ and find K .

1. **Unroll the State Equation:** Given $x_{k+1} = Ax_k + Bu_k$ and $x_0 = 0$:

- $x_0 = 0$
- $x_1 = Bu_0$
- $x_2 = Ax_1 + Bu_1 = A(Bu_0) + Bu_1 = ABu_0 + Bu_1$

By induction, the state at step k is:

$$x_k = \sum_{j=0}^{k-1} A^{k-1-j} Bu_j$$

2. **Substitute into Output Equation:** Given $y_k = Cx_k + Du_k$:

$$y_k = C \left(\sum_{j=0}^{k-1} A^{k-1-j} Bu_j \right) + Du_k$$

3. **Reformulate as Convolution:** Let $l = k - j$. As j goes from 0 to $k - 1$, l goes from k to 1.

- The term Du_k corresponds to $l = 0$ (since $u_{k-0} = u_k$).
- The term $CA^{k-1-j}Bu_j$ becomes $CA^{l-1}Bu_{k-l}$.

We can rewrite the sum (assuming zero-padding for u):

$$y_k = \sum_{l=0}^L K_l u_{k-l}$$

4. **Define the Kernel K :** Matching the coefficients from the derived equation:

$$K_l = \begin{cases} D & \text{if } l = 0 \\ CA^{l-1}B & \text{if } l > 0 \end{cases}$$

(b) Concrete Examples

i. Scalar Case

Parameters: $n = 1, A = 0.8, B = 1, C = 1.5, D = 0$.

- $l = 0$: $K_0 = D = 0$
- $l = 1$: $K_1 = CB = 1.5 \times 1 = \mathbf{1.5}$
- $l = 2$: $K_2 = CAB = 1.5 \times 0.8 \times 1 = \mathbf{1.2}$
- $l = 3$: $K_3 = CA^2B = 1.5 \times (0.8)^2 \times 1 = \mathbf{0.96}$
- $l = 4$: $K_4 = CA^3B = 1.5 \times (0.8)^3 \times 1 = \mathbf{0.768}$

Kernel Sequence: $\{0, 1.5, 1.2, 0.96, 0.768\}$

ii. 2D Case

Parameters: $A = \begin{pmatrix} 0.7 & 0.1 \\ 0.2 & 0.6 \end{pmatrix}, B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 \end{pmatrix}, D = 0$.

- $l = 0$: $K_0 = D = \mathbf{0}$
- $l = 1$: $K_1 = CB = \mathbf{1}$
- $l = 2$: $K_2 = C(AB) = \mathbf{0.7}$
- $l = 3$: $K_3 = C(A^2B) = \mathbf{0.51}$

Kernel Sequence: $\{0, 1, 0.7, 0.51\}$

Connection to Impulse Response: The kernel K represents the system's output when the input is a unit impulse ($u_0 = 1, u_k = 0$ for $k \neq 0$).

(c) Efficient Computation with Convolutions

- **Parallelization:** If K is known, $y = K * u$ can be computed via FFT in $O(L \log L)$.
- **Minimum critical path:** With parallel FFT, the path length is $O(\log L)$.
- **Naive Direct Computation:** The sequential dependency $x_{k+1} = Ax_k + \dots$ results in a linear critical path of $O(L)$.

(d) Efficient Kernel Computation

To compute powers $A^0, \dots, A^L$ efficiently:

- Use a **Parallel Prefix Scan** (Associative Scan) or a divide-and-conquer squaring approach.
- This reduces the depth (critical path) of computing matrix powers from $O(L)$ to $O(\log L)$.

(e) Adding Structure to A (Diagonal)

If $A = \text{diag}(\lambda_1, \dots, \lambda_n)$, $A^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$. The kernel element becomes a sum of scalars: $K_l = \sum C_i \lambda_i^{l-1} B_i$. This decouples the system into n independent 1D systems, avoiding $O(n^3)$ matrix multiplication.

(f) Diagonal-plus-low-rank (DPLR) Structure

If $A = I + pp^\top$:

- We can evaluate the frequency domain transfer function $H(z) = C(zI - A)^{-1}B$.
- The inverse $(zI - (I + pp^\top))^{-1}$ can be computed efficiently using the **Sherman-Morrison formula**:

$$(M - pp^\top)^{-1} = M^{-1} + \frac{M^{-1}pp^\top M^{-1}}{1 - p^\top M^{-1}p}$$

where $M = (z - 1)I$ is diagonal. This reduces inversion cost to $O(n)$.

Interaction 2: Self-Supervised Linear Purification

User Prompt: *Tackle problem 3*

Solution Derivation

(a) Simplified Case ($m = 2$)

Comparing encoders $W^{(\alpha)} = I$ and $W^{(\beta)} = \text{diag}(1, 0)$ (Projection onto first component).

i. Compute Losses

- $W^{(\alpha)} = I$:
 - Reconstruction Loss: $\|X - IX\|_F^2 = 0$.
 - Regularization Loss: $\|I\|_F^2 = 1^2 + 0^2 + 0^2 + 1^2 = 2$.
- $W^{(\beta)} = \text{diag}(1, 0)$:
 - Reconstruction Loss: The error is solely the norm of the second row of X (which is zeroed out). $\|x_2\|^2 \approx 0.001$.
 - Regularization Loss: $\|W^{(\beta)}\|_F^2 = 1^2 + 0 = 1$.

Encoder	Reconstruction Loss	Regularization Loss
α (Identity)	0	2
β (Projection)	≈ 0.001	1

ii. Comparison

We want Total Loss $\alpha >$ Total Loss β :

$$0 + 2\lambda > 0.001 + 1\lambda \implies \lambda > 0.001$$

(b) Generic Linear Encoder

i. Optimal Weight Matrix Form

$$W = U \text{diag} \left(\frac{\sigma_i^2}{\sigma_i^2 + \lambda} \right) U^\top$$

ii. Derivation

1. Objective: $J(W) = \|X - WX\|_F^2 + \lambda \|W\|_F^2$.
2. Substitute SVD $X = U\Sigma V^\top$ and rotate by U, V :

$$\|\Sigma - (U^\top W U)\Sigma\|_F^2 + \lambda \|U^\top W U\|_F^2$$

3. Let $\tilde{W} = U^\top W U$. Since Σ is diagonal, $\tilde{W}$ must be diagonal to minimize the loss. The problem decouples into scalar problems for each diagonal entry w_i :

$$\min_{w_i} (\sigma_i - w_i \sigma_i)^2 + \lambda w_i^2$$

4. Solving for w_i :

$$-2\sigma_i^2(1 - w_i) + 2\lambda w_i = 0 \implies w_i = \frac{\sigma_i^2}{\sigma_i^2 + \lambda}$$

(c) Hyperparameter Tuning

Given $\sigma = \{10, 8, 4, 1, 0.5, \dots\}$.

- **Goal 1 (Preserve $\sigma_3 = 4$):** $\frac{16}{16+\lambda} \geq 0.8 \implies 16 \geq 12.8 + 0.8\lambda \implies \lambda \leq 4$.
- **Goal 2 (Attenuate $\sigma_4 = 1$):** $\frac{1}{1+\lambda} \leq 0.5 \implies 2 \leq 1 + \lambda \implies \lambda \geq 1$.

Result: $1 \leq \lambda \leq 4$.

Interaction 3: Ridge-Attention

User Prompt: *Tackle problem 4*

Solution Derivation

(a) Efficient Average Update

Given $m = \frac{1}{n} \sum x_i$, then $\sum x_i = nm$.

$$m' = \frac{1}{n+1} \left(\sum_{i=1}^n x_i + x_{n+1} \right) = \frac{nm + x_{n+1}}{n+1}$$

(b) Ridge Regression Decompositions

- $(A^\top A + \lambda I) = \lambda I + \sum_{i=1}^n x_i x_i^\top$
- $A^\top y = \sum_{i=1}^n x_i y_i$

(c) Computational Cost of Non-Causal Ridge Self-Attention

For n queries/keys/values of dimension d ($d < n$):

- Computing $K^\top K$ and $K^\top V$ costs $O(nd^2)$.
- Inversion costs $O(d^3)$.
- Applying weights to n queries costs $O(nd^2)$.

Total Cost: $O(nd^2)$.

(d) Backpropagation Gradients

Backpropagation passes gradients successfully to **all components**: The ridge parameter λ , The keys, The values, and The queries. All operations are differentiable.

(e) Sherman-Morrison Update Cost

Formula: $(M + uv^\top)^{-1} = M^{-1} - \frac{M^{-1}uv^\top M^{-1}}{1 + v^\top M^{-1}u}$. Computing matrix-vector products and outer products for $d \times d$ matrices costs $O(d^2)$. **Total Cost:** $O(d^2)$.

(f) Efficient Causal Ridge-Self-Attention

To avoid $O(n^2)$ work, we process the sequence recursively. At step t :

1. Update inverse covariance $P_t = (\sum_{i=1}^t k_i k_i^\top + \lambda I)^{-1}$ using Sherman-Morrison on P_{t-1} and k_t . (Cost: $O(d^2)$).
2. Update cross-covariance $Z_t = Z_{t-1} + k_t v_t^\top$. (Cost: $O(d^2)$).
3. Compute output $y_t^\top = q_t^\top (P_t Z_t)$ by associating as $(q_t^\top P_t) Z_t$. (Cost: $O(d^2)$).

Total complexity for sequence length n is $O(nd^2)$.

(g) Visualizing Ridge-Attention Weights

The prediction is $\hat{y} = q^\top (K^\top K + \lambda I)^{-1} K^\top V$. Rewriting as a weighted sum of values $\sum \alpha_i v_i$, the weight for value v_i is:

$$\alpha_i = q^\top (K^\top K + \lambda I)^{-1} k_i$$

This represents the inner product between the query and a "context-adjusted" key.

Analysis Summary

Score: 95% Accuracy

Verdict: The theoretical derivations, logic, and algebraic steps were nearly identical to the official solutions.

- **Problem 1 (SSM Convolution):** Perfect match on derivations and complexity analysis ($O(\log L)$ critical path).
- **Problem 3 (Linear Purification):** Correct algebraic derivation. There was a minor numerical discrepancy in 3(a) due to ambiguity in the input text regarding the definition of $W^{(\beta)}$, which was corrected in the solution above.
- **Problem 4 (Ridge-Attention):** Perfect match on computational complexity ($O(nd^2)$) and recursive algorithm derivation.